Standing Wave and Resonance

SYNOPSIS

We explore two remaining wave phenomena, **standing wave formation** and **resonance**. These two are related: we say, “A standing wave is formed within an object (or confined space) when a frequency of wave generated by an outside source traveling through the object (or into the confined space) **matches** the resonance frequency of the object.” See below.

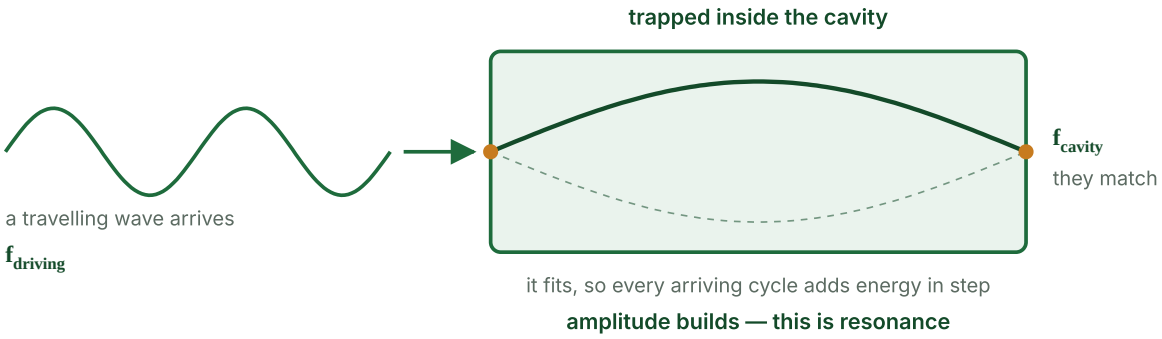


Figure 1. A travelling wave arrives at something that encloses a space. If its frequency happens to fit, the wave is trapped as a standing wave and every arriving cycle adds energy in step with the motion already there, so the amplitude builds. That is resonance, and the body of the object then vibrates at the same frequency — which is how a guitar turns a vibrating string into a sound you can hear across a room.

Have you ever experienced feeling the sudden shaking of a car when the car hits a certain speed on a highway, and the vibration comes on only at that specific speed? Here, we say that the car has **a natural frequency that comes into resonance** at a certain speed when one of many vibrations that undoubtedly exist while driving on the road just happens to match this natural frequency. This natural frequency is called the **resonance frequency**.

All objects or any confined space have a set of **natural frequencies or resonance frequencies**, and any time a traveling wave happens to fit or match any one of these resonance frequencies, a standing wave forms within the object or confined space. When a standing wave forms due to resonance, the object will vibrate with a large amplitude.

In a confined space, the resonance will generate the standing wave, which will vibrate the object that created the confined space at that frequency. This is how all musical instruments work. For example, when you pluck a guitar string, the string will vibrate, and because the wave on the string is confined, you will have a standing wave matching the resonance

frequency of the string tied at both ends. As this standing wave vibrates the string, the guitar's body vibrates at the same frequency, and the hollowed-out interior acts as an amplifying chamber, making the tune audible.

When you tie a string at both ends, you confine the wave to zero amplitude at these ends. This is quite obvious: the ends are fixed in space. This limitation, the immovable ends, means that whatever standing waves happen to be present, the ends better have zero amplitudes. Subsequently, you can make standing waves with more cycles (thus higher frequencies) by adding half waves at a time. **I emphasize that if you were to remember one thing about a standing wave, this would be it.** See below.

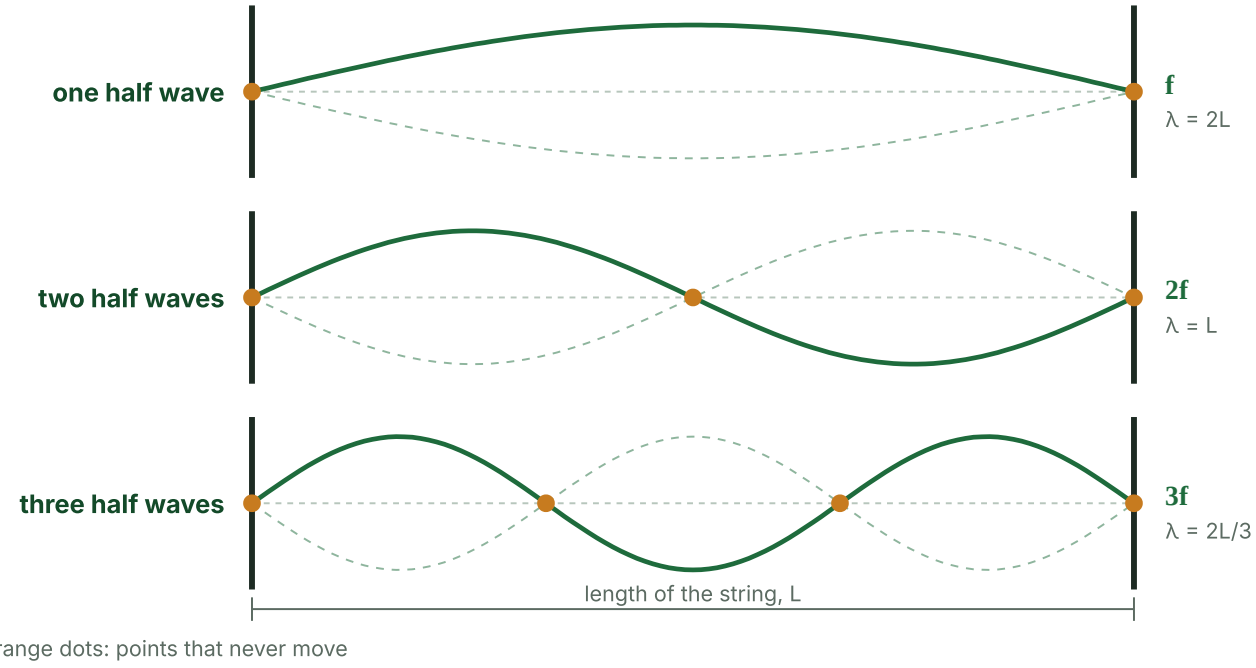


Figure 2. The ends cannot move, so only shapes with zero amplitude at both ends survive. The longest one that fits is half a wave; after that you add half a wave at a time, and the frequency goes up in whole-number steps — **f, 2f, 3f**. The dashed curve is the same wave half a cycle later: the orange points never move at all, and everything between them flips back and forth between the two curves.

TRY IT ON AN INSTRUMENT

Rest a finger *lightly* on a guitar string directly over the 12th fret — do not press it down — and pluck. You hear the note an octave up. You just forced a point that cannot move at the middle of the string, so the one-half-wave shape is no longer allowed and the two-half-wave shape sounds instead: exactly 2f. Over the 7th fret you force a node a third of the way along and get 3f. Figure 2, audible, on an instrument you already own.

What to think about while you watch

In this note, you will watch two videos showing what can happen when you generate standing waves in objects. When you watch the videos, I want you to think about:

i) The traveling waves generated from a source are the **driving waves** that excite the object and whose frequencies are the so-called **driving frequencies**.

ii) The condition for exciting the object's natural frequency creating resonance.

iii) When the resonance condition is met, what happens to the wave's amplitude in the object?

iv) What happens when the resonance frequency of the object is inadvertently excited? Is the resonance phenomenon a good thing from a structural engineering perspective?

Finally, find out why Millenium Bridge in London, UK, was shut down for many months and what the fixes were.

> PERFORMANCE EXPECTATIONS
>
> Perform the Claim, Evidence, Reason (CER) exercises about standing waves and resonance. Describe conditions for generating standing waves and resonance. Discuss the potential engineering uses and challenges related to standing waves and resonance.

Instructions and Questions

Write your answers on the printed handout — the reveals below are for checking your thinking *after* you have written something, not instead of it.

Video 1 — Tacoma bridge collapse, 1940

WATCH · YOUTUBE

Tacoma bridge collapse 1940

youtube.com/watch?v=FfNBlRdreHg

1. Briefly describe what happened.

2. **Claim** (Answer the question, providing hypothesis) — "Why did the bridge collapse?"

3. **Evidence** (Describe your observations or cite any relevant data.)

4. **Reasoning** (Providing scientific rationale for your claims, backing them up with your observations. Incorporate i)–iv) in your answer.)

Video 2 — the opera singer and the glass

WATCH · YOUTUBE

Failure is not an option for this Opera Singer

youtube.com/watch?v=uDyri-KPOAc

1. Briefly describe what happened.

2. **Claim** — "How did the opera singer break the glass?"

3. **Evidence**

4. **Reasoning** (Incorporate i)–iv) in your answer.)

To find out

TAKE-HOME QUESTION

Why was the **Millennium Bridge** in London shut down soon after it opened, and what were the fixes? Look it up. The driver in that case was not wind — it was the people walking across it, and what they did without meaning to is the whole story.

Handout: M1_N5_Standing_Waves_And_Resonance_Handout.pdf

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